

Vol 2 Chapter 9 — The Higgs Sector

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Chapter 9 — The Higgs Sector

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

The Higgs boson, discovered at the LHC in 2012, completes the Standard Model. Its mass is $m_H \approx 125.10$ GeV; its vacuum expectation value $v \approx 246.22$ GeV sets the electroweak scale; the quartic self-coupling λ_H and the Yukawa couplings to fermions are the Higgs sector's parameters. In the Standard Model, all of these are inputs.

BST derives the Higgs mass at 0.07-0.11% via two independent substrate routes (D-tier dual-route per Elie's Vol 2 Ch 9 v0.3 work and Grace's INV-4833 catalog entry). The full mechanism chain — why the Higgs has its specific vacuum-condensation structure — is at PARTIAL DERIVED status, with multi-month research closing the remaining substrate-mechanism pieces.

9.1 The Higgs mass at two routes

Route A (Elie's substrate-mechanism reading): m_H emerges from Casimir-weighted substrate vacuum scaling at the substrate's natural Higgs K-type, giving $m_H \approx 125.0$ GeV at 0.07%.

Route B (Grace's INV-4833 catalog reading): m_H emerges as a specific BST-primary integer combination, with the substrate-mechanism reading via $SU(2)$ doublet condensation in the canonical Bergman anchor, giving the same m_H at 0.11%.

The two routes are independent and agree at sub-percent precision. The Quaker discipline tier label is PARTIAL DERIVED because the full mechanism chain — particularly the substrate-side derivation of the Higgs vacuum structure that the routes assume — is multi-month research. The match precisions are at D-tier; the mechanism rigor is one tier below.

9.2 The vacuum expectation value v

The electroweak vacuum expectation value $v = 246.22$ GeV is in standard physics a fundamental input. In BST, v emerges from substrate Casimir scaling: v is the substrate's natural condensation scale for the Higgs K-type, related to the substrate's Bergman normalization $c_{FK} \cdot \pi^{9/2} = 225$ through specific mechanism chains.

The substrate-derivation of v is at partial-derived status (multi-month). The numerical value matches experiment at sub-percent when the BST-primary combination is identified.

9.3 The quartic coupling λ_H

λ_H controls the Higgs self-interaction strength. Substrate-derivation gives $\lambda_H \approx 0.13$ at the percent level, consistent with experiment. The mechanism is the substrate's Casimir-eigenvalue weighting on the Higgs K-type quartic interaction.

9.4 Yukawa couplings

The Higgs couples to fermions via Yukawa interactions, with coupling strengths proportional to the fermion masses. In BST, the Yukawa-coupling structure is fixed by the substrate's K-type weight spectrum on fermion generations — the same structure that gives the fermion masses themselves (Chapters 3 and 5).

The Yukawa ratio decoupling (Lyra T2450 work, Friday May 22, 2026) addresses the structural relationships among the Yukawa couplings across the three fermion generations. The substrate-derivation pins these ratios at the percent level; multi-month research is refining the precision.

9.5 The Higgs mechanism on the substrate

Standard electroweak theory's Higgs mechanism — the spontaneous symmetry breaking of $SU(2) \times U(1) \rightarrow U(1)_{em}$ when the Higgs field acquires a vacuum expectation value — has a substrate-side reading via the bundle-connection framework of Lyra T2477 (Vol 1 Ch 8 §8.6). The substrate's $SO(2)$ factor and the substrate's $SU(2)$ rank-doublet structure undergo Bergman-bundle deformation when the substrate vacuum is in its Higgs-condensed configuration, with the residual $U(1)_{em}$ corresponding to the unbroken combination.

The full substrate-side Higgs-mechanism derivation is the multi-month rail that closes the PARTIAL DERIVED status on this chapter. When closed, the Higgs sector will join the rest of the volume's D-tier ratified derivations.

9.6 What comes next

Chapter 10 treats neutrinos with the seesaw mechanism anchored at BST primary 17.

Where to look this up: Higgs mass dual route: Elie Vol 2 Ch 9 v0.3 + Grace INV-4833. Yukawa ratio decoupling: Lyra T2450. Higgs as Bergman-bundle deformation: T2477. For standard Higgs-sector treatments: Peskin–Schroeder Chapters 20–21; Djouadi 2008 review *Physics Reports* 457:1.